

Name: _____

Class: _____

Date: _____

Unit 5 Review Packet

Impact of Computing



How to use this packet

Match the vocabulary, then work the mixed review questions. These are the exact words and question types you'll see on the unit test and the AP exam.

Vocab Match 1

- _____ 1. **Computing innovation**
- _____ 2. **Beneficial effect**
- _____ 3. **Harmful effect**
- _____ 4. **Unintended consequence**
- _____ 5. **Responsible programming**
- _____ 6. **Digital divide**
- _____ 7. **Socioeconomic**
- _____ 8. **Geographic**
- _____ 9. **Demographic**
- _____ 10. **Equity**

- A. fairness — everyone gets what they need to succeed
- B. related to income, education, and social class
- C. an outcome that hurts people, groups, or society
- D. related to location (rural vs. urban, country)
- E. an effect that the creators did not plan for
- F. an outcome that helps people or society
- G. related to characteristics like age, race, or gender
- H. includes a program as an integral part of its function (physical, nonphysical software, or a concept)
- I. considering the unintended uses and effects of a program before and after release
- J. differing access to computing devices and the Internet

Vocab Match 2

- _____ 1. **Bias**
- _____ 2. **Algorithmic bias**
- _____ 3. **Training data**
- _____ 4. **Machine learning**
- _____ 5. **Representative data**
- _____ 6. **Crowdsourcing**
- _____ 7. **Citizen science**
- _____ 8. **Crowdfunding**
- _____ 9. **Public data**
- _____ 10. **Collaboration**

- A. raising money from many people online
- B. research done with help from non-scientists who contribute data
- C. data that reflects the whole population it will be used on
- D. bias that results from an algorithm's design or its data
- E. data made freely available for anyone to use
- F. unfair preference for or against a group
- G. getting input or information from many people via the Internet
- H. the data a machine learning model learns from
- I. people working together toward a common goal
- J. computers learning patterns from data instead of explicit rules

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Vocabulary (continued)

Vocab Match 3

- _____ 1. Intellectual property
- _____ 2. Copyright
- _____ 3. Creative Commons
- _____ 4. Open source
- _____ 5. Plagiarism
- _____ 6. Encryption
- _____ 7. Symmetric key
- _____ 8. Public key
- _____ 9. Certificate authority
- _____ 10. Malware

- A. using someone else's work without permission and presenting it as your own
- B. pairs a public key for encryption with a private key for decryption
- C. the process of encoding data to prevent unauthorized access
- D. legal protection that gives creators control over how their work is used
- E. issues digital certificates that validate ownership of encryption keys (a trust model)
- F. a license creators use to give others the right to share, use, and build upon their work
- G. creations of the mind — code, art, music, writing — owned by the creator
- H. encryption that uses one key for both encrypting and decrypting
- I. code that can be freely shared and modified
- J. software intended to damage a computing system or take partial control of it

Unit 5 Review Packet

Mixed Review

Mixed Review

1. Which is an example of **multifactor authentication**?

- A. A long password
- B. Two different passwords
- C. A password plus a code texted to your phone
- D. A security question

2. Which action would MOST directly reduce the digital divide?

- A. Providing free Wi-Fi hotspots to low-income families
- B. Releasing a more expensive phone
- C. Requiring all forms to be completed online
- D. Increasing the price of broadband

3. A college admissions algorithm does not use race as an input, but it does use ZIP code. Why might it still be biased?

- A. ZIP codes are numbers
- B. ZIP codes change every year
- C. Algorithms can't use location data
- D. ZIP codes can be strongly correlated with race and income, acting as a proxy

4. A school requires all homework to be completed on a platform that needs high-speed video streaming. Which students are MOST likely disadvantaged?

- A. Students in rural areas with limited broadband
- B. Students with new laptops
- C. Students who type quickly
- D. Students in after-school clubs

5. Which is an example of crowdsourcing?

- A. A company's 3 engineers design a new app
- B. A computer running a sorting algorithm
- C. A teacher grading tests alone
- D. An online encyclopedia written and edited by thousands of volunteers

6. Which TWO are benefits of crowdsourcing? (*Select two.*)

- A. Many participants can solve problems too large for a small team.
- B. It guarantees all data is accurate.
- C. It can reduce the cost of collecting data.
- D. It eliminates the need for any experts.

7. What is open source software?

- A. Software that is always free of bugs
- B. Programs made freely available that may be freely shared and modified
- C. Software owned by the government
- D. Software that can only be used offline

8. A traffic app relies on drivers to report accidents. Which is a potential drawback?

- A. Drivers cannot use apps
- B. The app cannot use GPS
- C. Crowdsourcing always costs more than hiring staff
- D. Reports may be inaccurate, and areas with few users get less coverage

Unit 5 Review Packet

Mixed Review

Mixed Review

9. Which TWO actions are most responsible for developers? (*Select two.*)

- A. Considering how an app could be misused before release
- B. Assuming users will only use the app as intended
- C. Testing with diverse groups of users to discover potential harms
- D. Avoiding documentation to protect the code

10. Which is MOST likely to lead to bias in a facial recognition system?

- A. Using a fast processor
- B. Storing images in a compressed format
- C. Training it mostly on images of one demographic group
- D. Writing comments in the code

11. Alice wants to send Bob a message using public key encryption. Which key should she use to encrypt it?

- A. Bob's public key
- B. Alice's private key
- C. Bob's private key
- D. Any symmetric key

12. Which is an example of a computing innovation being used in a way its creators did not originally intend?

- A. The World Wide Web, built for scientists, now used for shopping and social media
- B. A calculator used for addition
- C. A word processor used to type an essay
- D. A GPS used for directions

13. Which TWO statements are true? (*Select two.*)

- A. The digital divide exists only between countries.
- B. The digital divide can be affected by government actions.
- C. The digital divide includes differences in access based on location.
- D. Smartphones have completely eliminated the digital divide.

14. A photographer wants anyone to use her photos as long as they give her credit. Which is the best option?

- A. Keep full copyright and allow no use
- B. Delete the photos' metadata
- C. Use a Creative Commons Attribution (BY) license
- D. Use lossy compression

15. Which statement is true about the effects of computing innovations?

- A. Every effect is predicted before release
- B. Innovations only have beneficial effects
- C. An effect can be beneficial to some people and harmful to others
- D. Harmful effects only come from bad programmers

16. What is the best way to reduce bias in a machine learning model?

- A. Collect more data from the same source
- B. Remove all documentation
- C. Train and test it on data that represents a diverse population
- D. Run it on more computers

Unit 5 Review Packet

Mixed Review

Mixed Review

17. Which TWO statements are true? (*Select two.*)

- A. Bias can be introduced by the data used to train an algorithm.
- B. Algorithms are objective, so they cannot be biased.
- C. Programmers can take actions to reduce bias in algorithms.
- D. Collecting more data always removes bias.

18. Which TWO are risks of sharing PII online? (*Select two.*)

- A. It can be combined with other data to build a profile of you.
- B. It is easy to permanently delete.
- C. It can be used for identity theft.
- D. Only the person who posted it can ever access it.

19. Which TWO statements are true? (*Select two.*)

- A. Material created on a computer is the intellectual property of its creator.
- B. Anything posted on the Internet is free to use for any purpose.
- C. Creative Commons licenses let creators allow sharing while keeping some rights.
- D. Open access means anyone can claim authorship of the research.

20. A student uses code from an open-source library in their Create task. What is the most appropriate action?

- A. Use it and present it as their own original work
- B. Avoid all outside code completely
- C. Use it and acknowledge the source in their program and written responses
- D. Rename the variables so it isn't recognized

21. What is the digital divide?

- A. The difference between hardware and software
- B. The gap between binary and decimal numbers
- C. Differing access to computing devices and the Internet based on socioeconomic, geographic, or demographic characteristics
- D. Splitting a program into modules

22. A fitness app publishes a public heat map of users' running routes. The map accidentally reveals the locations of secret military bases. Which best describes this?

- A. An intended beneficial effect
- B. A harmful, unintended consequence of a feature meant to be beneficial
- C. A syntax error
- D. A beneficial effect for everyone

23. Which is an example of citizen science?

- A. Volunteers record backyard bird sightings in an app that researchers analyze
- B. A scientist writes a paper alone
- C. A company sells science kits online
- D. A student watches a science video

24. What is the role of a certificate authority?

- A. It writes computer viruses
- B. It decrypts all Internet traffic
- C. It stores everyone's passwords
- D. It issues digital certificates that validate ownership of encryption keys

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Challenge Questions

Challenge 5.1

Choose a computing innovation (it can be your Create task idea). Describe one beneficial effect and one harmful effect, and explain one specific way the harmful effect could be reduced.

Challenge 5.2

Some people argue that because most people own smartphones, the digital divide is basically solved. Evaluate this argument using at least two specific points.

Challenge 5.3

Connect Topic 1.1 (collaboration) to this topic: explain how a development team with diverse backgrounds and perspectives could reduce bias in a computing innovation. Give a specific example.

Challenge 5.4

Explain how the digital divide (Topic 5.2) could affect the results of a crowdsourced project. Use the pothole app or another example.
